

Translated

From: e-maxx.ru

Check if point belongs to the convex polygon in $O(\log N)$

Consider the following problem: you are given a convex polygon with integer vertices and a lot of queries. Each query is a point, for which we should determine whether it lies inside or on the boundary of the polygon or not. Suppose the polygon is ordered counter-clockwise. We will answer each query in $O(\log n)$ online.

Algorithm

Let's pick the point with the smallest x-coordinate. If there are several of them, we pick the one with the smallest y-coordinate. Let's denote it as p_0 . Now all other points $p_1, \dots, p_n$ of the polygon are ordered by their polar angle from the chosen point (because the polygon is ordered counter-clockwise).

If the point belongs to the polygon, it belongs to some triangle p_0, p_i, p_{i+1} (maybe more than one if it lies on the boundary of triangles). Consider the triangle p_0, p_i, p_{i+1} such that p belongs to this triangle and i is maximum among all such triangles.

There is one special case. p lies on the segment (p_0, p_n) . This case we will check separately. Otherwise all points p_j with $j \leq i$ are counter-clockwise from p with respect to p_0 , and all other points are not counter-clockwise from p . This means that we can apply binary search for the point p_i , such that p_i is not counter-clockwise from p with respect to p_0 , and i is maximum among all such points. And afterwards we check if the point is actually in the determined triangle.

The sign of $(a - c) \times (b - c)$ will tell us, if the point a is clockwise or counter-clockwise from the point b with respect to the point c . If $(a - c) \times (b - c) > 0$, then the point a is to the right of the vector going from c to b , which means clockwise from b with respect to c . And if $(a - c) \times (b - c) < 0$, then the point is to the left, or counter clockwise. And it is exactly on the line between the points b and c .

Back to the algorithm: Consider a query point p . Firstly, we must check if the point lies between p_1 and p_n . Otherwise we already know that it cannot be part of the polygon. This can be done by checking if the cross product $(p_1 - p_0) \times (p - p_0)$ is zero or has the same sign with $(p_1 - p_0) \times (p_n - p_0)$, and $(p_n - p_0) \times (p - p_0)$ is zero or